2/03

Last time we talked about a few common probability distributions: bernoulli, binomial, and Normal.

last time
* bernoulli
* binomial
* normal

Bernoulli is for binary outcomes.

bernoulli: $X \in \{0, 1\}$

and it is parameterized by the probability of the $X=1$ outcome (success), "p".

$\mathbb{P}(X=1) = p \quad \mathbb{E}(X) = p \quad sd(X) = \sqrt{p(1-p)}$

We worked out that its expected value is p, and standard deviation is $\sqrt{p(1-p)}$.

The binomial distribution arises when we have a sequence of independent bernoulli trials, all with the same success probability p. The total # of successes over that collection of n binary events is a binomial.

binom: $X_1, \ldots, X_n \overset{iid}{\sim} Bern(p)$

$Y = \sum_{i=1}^{n} X_i \sim Binom(n, p)$

Its expected value is n times p, and standard deviation is $\sqrt{np(1-p)}$.

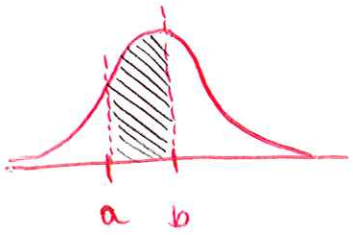
$\mathbb{E}(Y) = np \quad var(Y) = np(1-p)$

$sd(Y) = \sqrt{np(1-p)}$

The other probability distribution we talked about was the Normal, aka Gaussian, distribution. This is a continuous distribution. Bernoulli takes binary values, binomial takes integer values from 0 to n, and Normal RVs take any value on the real number line $(-\infty, \infty)$.

Normal (gaussian) * continuous * $X \in (-\infty, +\infty)$

If we visualize the normal distribution, its probability density function is a smooth curve.



shaded area = $\mathbb{P}(a<X<b)$

The probability of a normal variable taking a value in a certain interval is given by the corresponding area under the pdf curve.

The normal dist. is a family of distributions parameterized by expected value, aka mean, $\mathbb{E}(X)=\mu$, and its variance, $\text{var}(X)=\sigma^2$.

$$\mathbb{E}(X)=\mu \qquad \text{var}(X)=\sigma^2$$

The normal dist. has some special properties:

1. Linear transformations of a normal RV are still Normal.

$$X \sim N(\mu,\sigma^2) \Rightarrow aX+b \sim N(a\mu+b, a^2\sigma^2)$$

2. Sums of independent Normal RVs are also normal.

$$X \sim N(\mu_X,\sigma_X^2) \qquad Y \sim N(\mu_Y,\sigma_Y^2) \quad \text{and} \quad X \perp\!\!\!\perp Y \text{ (indep)}$$

$$\Rightarrow X+Y \sim N(\mu_X+\mu_Y, \sigma_X^2+\sigma_Y^2)$$

Their means and variances both add.

These distributions quantify the likelihood of different possible outcomes we could observe.

So far, the type of problem we've talked about is this:

We start with a probability model, and reason about what kind of data we would be likely to see.

start: probability model — want: what data am I likely to see?

For example:

15% of population left-handed,
LH out of 10 ~ Binom(10, 0.15)

$\mathbb{P}$(≥ 2 people LH in sample of 10) = ?

often, we want to reason in the opposite direction. We observe data, and want to know something about the population distribution.

start: observed data — want: population probability model

For example:

- collect data
- 1 LH person out of 10

→ what % of Americans are left handed?

To answer that question in a mathematically rigorous way, we can frame this as a question about a probability model. If the true percentage of people who are left handed is p, if we randomly select a person from the population, the prob. they are LH is p.

p = % LH in whole population

If we randomly select 10 people, the # of LH people in that

sample follows a Binomial(10, p) distribution.

LH people in random sample of 10 ~ Binom(10, p)

We don't know p. The actual data observed was 1 LH person out of 10. So based on our knowledge of the binomial dist, we could ask: what values of p are plausible?

observe Y=1 , plausible values of p?

To understand why and how we can use sample data to learn things about an overall population, it's important to understand some key statistical properties of sample data.

if we observe random samples of some quantity, the sample average doesn't necessarily equal the average of the broader population dist, but it will be close.

sample avg ≠ population avg

How close? It depends on how much data we collect, i.e. how many samples.

how close are they?

↳ depends on n

Suppose overall blood pressure of adults in the US follows a normal dist w/mean = 120 and sd = 10.

X = BP $\quad$ $X \sim N(120, 10^2)$

If we go out and measure one random person's blood pressure, it could be near 120, or above or below.

single observed X: could be > 130

If we randomly select 5 people, measure their BPs and take the average, could the avg be over 130? Maybe, but it would require us to have chosen 5 people who mostly have high BP, by random chance.

5 people observed: avg BP > 130?
if we get all people w/high BP by chance

But what if we selected 1000 people? If we sample them randomly, it would be hard to get 1000 people where most of them have high BP just by coincidence.

1000 people observed: avg BP > 130?
unlikely, huge coincidence

Using simulated data helps illustrate this point.

~ examples in notebook ~

larger sample → sample avg closer to population mean

As we collect larger and larger random samples, the sample avg tends to be closer and closer to the overall average of the population we're sampling from.

Formally, this is called the **Law of Large Numbers.**

If X is some quantity that follows a population dist with mean $E(X) = \mu$, then if we observe random samples of X, the sample average approaches the pop. avg μ as the sample size n increases.

$X \sim p(x)$, $E(X) = \mu$ $\quad \frac{X_1 + X_2 + \dots + X_n}{n} \to \mu$ as $n \to \infty$

Another example of this involves **sample proportions.**

If the overall percentage of left-handed adults in the US is 15%, the # of people in a random sample of size n who are left-handed is $Y \sim \text{Binom}(n, 0.15)$.

15% of population left handed
Y = # LH in sample of n
$Y \sim \text{Binom}(n, 0.15)$

The expected number is $\mathbb{E}(Y) = np = n * 0.15$.

$\mathbb{E}(Y) = np = 0.15n$
↳ expect 15% LH in sample

we expect 15% of people in the sample to be LH.

$\mathbb{E}\left(\frac{Y}{n}\right) = \frac{0.15n}{n} = 0.15$ define $\hat{p} = \frac{Y}{n}$

For a sample of 10 people, the actual observed fraction of LH people can be $\frac{0}{10}$, $\frac{1}{10}$, $\frac{2}{10}$, etc. Due to random chance, we could end up with 0 people, or 3 people, and it wouldn't be shocking.

$\mathbb{P}(\hat{p} = 0\%) = \mathbb{P}(Y = 0) \approx 0.20$

$\mathbb{P}(\hat{p} \geq 30\%) = \mathbb{P}(Y \geq 3) \approx 0.18$

If we sampled 100 people instead, we still expect 15% of them to be LH. But if we actually surveyed 100 people and zero of them were LH, that would be surprising. It would also be surprising if more than 30 of them were LH.

sample $n = 100$ instead

$\mathbb{P}(\hat{p} = 0) = \mathbb{P}(Y = 0) \approx 1$ in 10 mln

$\mathbb{P}(\hat{p} \geq 0.3) = \mathbb{P}(Y \geq 30) \approx 1$ in 10,000

with a larger sample size, it's less likely that we will get a sample proportion that is far away from the population proportion.

Next, let's discuss **sample means**.

In general, we have some quantity X with a population dist we'll generically refer to as $p(x)$. There is an expected value $\mathbb{E}(X) = \mu$.

$X \sim p(x)$ $\quad\quad$ $\mathbb{E}(X) = \mu$

LLN: $\bar{X}_n \to \mu$ as $n \to \infty$

The law of large numbers says that if the sample is very large, the sample mean tends to be close to the population mean μ.

What else can we say about the sample mean?

Just like X is a random variable with a distribution, $\bar{X}_n$ (the sample mean) is also a random variable with its own distribution.

$\bar{X}_n$ is random
↳ different values for each sample

Every time we go out and collect a random sample of data, the sample average is slightly different.

Let's continue thinking about the blood pressure example.

The blood pressure of adults in the US follows a population distribution that is normal w/mean 120 and sd = 10.

X = BP $\qquad X \sim N(120, 10^2)$

sample 5 people: $X_1, X_2, \ldots, X_5 \overset{iid}{\sim} N(120, 10^2)$

The BPs of 5 randomly selected people each come from the Normal$(120, 10^2)$ dist, and they are independent of each other.

Recall what we said about normal RVs: their sum is also normal (if independent, the means and variances add).

recall: sum of indep Normal RVs is also Normal

$(X_1 + \ldots + X_5) \sim N(120 + \ldots + 120, \quad 10^2 + \ldots + 10^2)$

The expected value of their sum is the sum of the expected values, and when they are independent, the variance of the sum is the sum of the variances.

From there, we can recognize that the sample mean equals that sum divided by 5, the # of observations.

$\bar{X} \sim N(?, ?)$

Dividing by a constant gives us another normal distribution, and we can determine its expected value and variance.

$$E(\bar{X}) = E\left(\frac{X_1 + \ldots + X_5}{5}\right) = \frac{1}{5} E(X_1 + \ldots + X_5) = \frac{1}{5}(5 * 120) = 120$$

$$\text{var}(\bar{X}) = \text{var}\left(\frac{X_1 + \ldots + X_5}{5}\right) = \frac{1}{5^2} \underbrace{(10^2 + \ldots + 10^2)}_{\text{var}(X_1 + \ldots + X_5)} = \frac{10^2}{5}$$

More generally, for the average of n data points, the sample avg has: 1) the same expected value as the population dist, but 2) smaller variance (decreasing as # samples increases).

generally: $\mathbb{E}(\bar{X}_n) = \mathbb{E}(X)$, $\text{var}(\bar{X}_n) = \frac{\text{var}(X)}{n}$

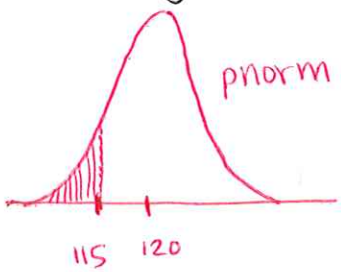
So in our blood pressure example, the average for 5 people is still normally distributed, with expected value and variance determined according to the usual rules.

BP example: $\bar{X}_5 \sim N(120, \frac{10^2}{5})$

Using that information, we can calculate the probability of the sample mean falling in a particular range.

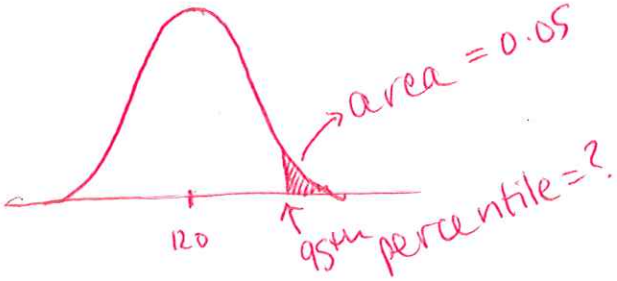
What is the prob. that the avg BP of 5 people is less than 115?

$\mathbb{P}(\bar{X}_5 < 115) = ?$



pnorm(115, mean=120, sd=$\frac{10}{\sqrt{5}}$)

What is the 95th percentile for the avg BP of 5 people? That is, what BP value will the 5-person-avg fall below 95% of the time?



qnorm(0.95, mean=120, sd=$\frac{10}{\sqrt{5}}$)

We found this normal distribution of the sample mean by reasoning that X itself was normal, and sums and averages of normal RVs are also normal.

$X \sim \text{Normal} \Rightarrow \bar{X}_n$ normal

what if X not normal?

It actually turns out that for large enough samples, the sample mean still approximately follows a Normal distribution (with that same expected value and variance). This is called the "central limit theorem."

central limit theorem

$X \sim p(x)$ not necessarily normal

$E(X) = \mu \qquad var(X) = \sigma^2$

$\bar{X}_n \approx N(\mu, \sigma^2/n)$ for large n

~ examples in notebook ~